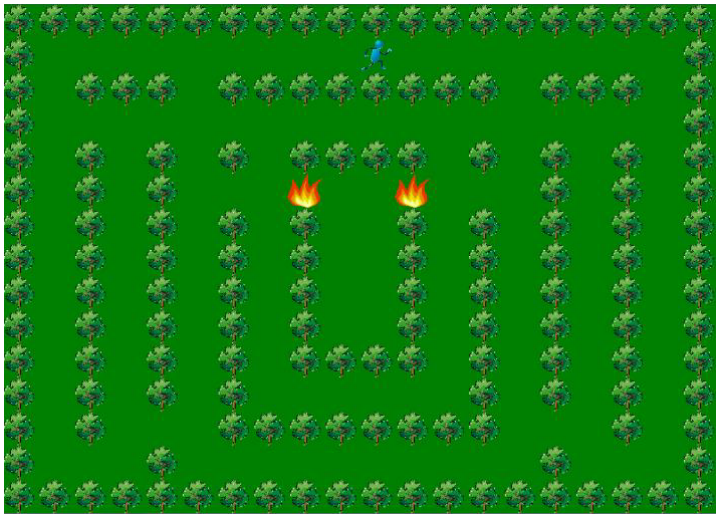
We can use the same grid to produce a more realistic world by using images instead of circles and rectangles:



Example Program

```
import pygame, sys
import random
from pygame.locals import *
pygame.init()
screen=pygame.display.set_mode((800,600))
blue=(0,0,255)
red=(255,0,0)
green=(0,128,0)
treeImg=pygame.image.load('tree.png').convert()
treeImg.set_colorkey(white)
treeImg=pygame.transform.scale(treeImg,(40,40))
fireImg=pygame.image.load('fire.png').convert()
fireImg.set_colorkey(white)
fireImg=pygame.transform.scale(fireImg,(40,40))

runnerImg=pygame.image.load('runner.png').convert()
runnerImg.set_colorkey(white)
runnerImg=pygame.transform.scale(runnerImg,(40,40))

def createMaze(): #set up the world as numbers
            #0-path of green
            #1-tree
            #2 - barrier of fire that could be removed

    row1= [1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1]#20 x 15 list
    row2= [1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1]
    row3= [1,0,1,1,1,0,1,1,1,1,1,1,1,1,0,1,1,1,0,1]
    row4= [1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1]
    row5= [1,0,1,0,1,0,1,0,1,1,1,1,0,1,0,1,0,1,0,1]
    row6= [1,0,1,0,1,0,0,0,2,0,0,2,0,0,0,1,0,1,0,1]
    row7= [1,0,1,0,1,0,1,0,1,0,0,1,0,1,0,1,0,1,0,1]
    row8= [1,0,1,0,1,0,1,0,1,0,0,1,0,1,0,1,0,1,0,1]
    row9= [1,0,1,0,1,0,1,0,1,0,0,1,0,1,0,1,0,1,0,1]
    row10=[1,0,1,0,1,0,1,0,1,0,0,1,0,1,0,1,0,1,0,1]
    row11=[1,0,1,0,1,0,1,0,1,1,1,1,0,1,0,1,0,1,0,1]
    row12=[1,0,1,0,1,0,1,0,0,0,0,0,0,1,0,1,0,1,0,1]
    row13=[1,0,1,0,0,0,1,1,1,1,1,1,1,0,0,0,1,0,1]
    row14=[1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,1,0,0,0,1]
    row15=[1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1]
    maze = [row1,row2,row3,row4,row5,row6,row7,row8,row9,row10,row11,row12,row13,row14,row15]
    return(maze)
```

Code continued on this page

```
def drawMaze(maze):
    for j in range(20):
        for i  in range(15):
            x=j*40
            y=i*40
            if maze[i][j] == 1 :
                screen.blit(treeImg,(x,y))
            elif maze[i][j] == 2:
                screen.blit(fireImg,(x,y))
            elif maze[i][j] == 0:
                pygame.draw.rect(screen,green,[x,y,40,40])

#==============================================================================
maze=createMaze()
row=1  #starting location of the player in the maze
column=10

while True:
    screen.fill(green)
    drawMaze(maze)
    screen.blit(runnerImg,(column*40,row*40))
    for event in pygame.event.get():
        if event.type==QUIT:
            pygame.quit()
            sys.exit()

#=============== player movement   to be added later==========================

     pygame.display.update()
```

We can now add player movement to have the player move in the maze.

```
#=============== player movement   ========================================
        if event.type==KEYDOWN:
            if event.key==K_UP and maze[row-1][column] == 0:
                row=row-1
            if event.key==K_DOWN and maze[row+1][column] == 0:
                row=row+1
            if event.key==K_LEFT and maze[row][column-1] == 0:
                column=column-1
            if event.key==K_RIGHT and maze[row][column+1] == 0:
                column=column+1
#===============end of player movement========================================
```